EGRE 245 Engineering Programming
Fall 2024
Mark Schwitzerlett

Lab Exercise #2: Total Cost To Paint A Wall

1 Overview

In this lab, you will write a program to calculate the cost of carpeting 3 rooms. Carpet is priced by the square foot. The total cost includes the price of the carpet, labor and sales tax.

Your program should

- Request the user to input the dimensions of the room in sq feet, wall height (`double`), wall width (`int`) and paint cost per gallon (`int`).
- Calculate and output the area of the wall in square feet.
- Calculuate and output the required number of cans of paint (one can of paint will cover 350 sq ft).
- Calculate and output the sales tax (4.5%) on for the paint.
- Calculate and output the total cost which includes the cost of the paint and sales tax.
- Any constants should be declared as `const` type within the body of your program NOT as directives.

2 Code Documentation

The very first lines in *all* of your programming project files this semester should be comments in the class header file (copy from header.c). Of course, the items in the square brackets [] in the comments should be filled in with the proper information.

In addition, the first output for all of your projects this semester, both for the lab exercises and the programming assignments, should be the name of the project and your name, e.g:

```
printf("Lab #2 - Joe Student\n");
```

Obviously, for the lab exercises, the output should include the names of both students in the lab group, e.g:

```
printf("Lab #2 - Joe Student and Jane Doe\n");
```

3 Sample Run

Your output should match the sample run below.

```
Lab #2: Mark Schwitzerlett and Suzy Q
EGRE 245 Engineering Programming Using C Lab #2 Cost to Paint a Wall

Input Wall Height  14.75

Input Wall Width  16

Input Paint Cost per Gallon  34.98


Wall area: 236.0 sq ft
Paint needed: 0.674 gallons
Cans needed: 1 can(s)
Paint cost: $34.98
Sales tax: $1.57
Total cost: $36.55


...Program finished with exit code 0
Press ENTER to exit console.
```

4 Deliverables

You should turn in a stand-alone, complete application program (your C source code, a single ".c" file) containing a `main()` function. All labs and projects this semester will be submitted in Gradescope (via a Canvas assignment).

Your file should be named Lab#2_FirstName_LastName.c

Due date: The lab is due at the end of your lab period, either Tuesday, Septemeber 9th or Thursday, September 11th.

Be sure to save a copy of your .c file.